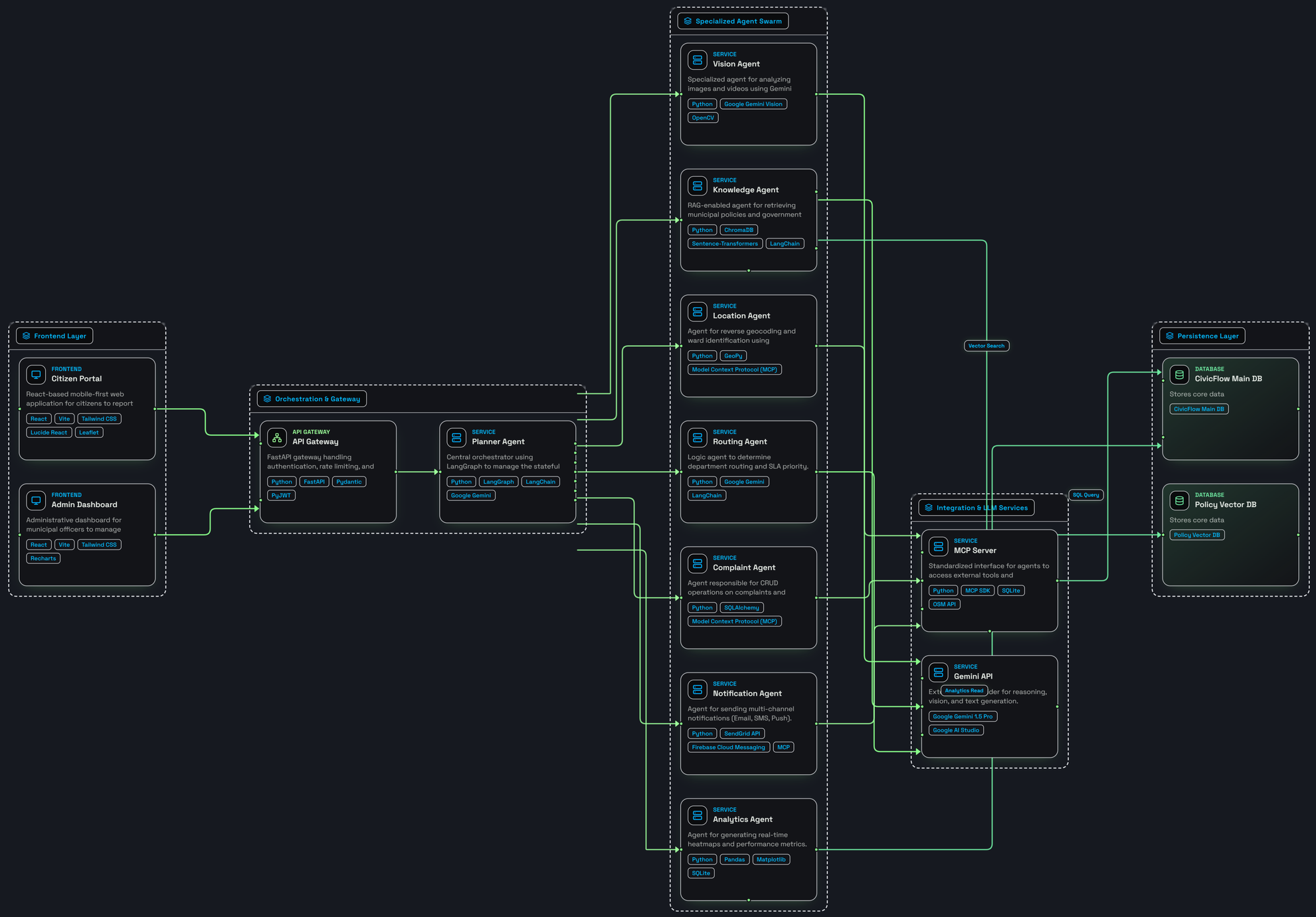
Frontend Layer
FRONTEND
Citizen Portal
React-based mobile-first web application for citizens to report
React
Vite
Tailwind CSS
Lucide React
Leaflet
FRONTEND
Admin Dashboard
Administrative dashboard for municipal officers to manage
React
Vite
Tailwind CSS
Recharts
Orchestration & Gateway
API GATEWAY
API Gateway
FastAPI gateway handling authentication, rate limiting, and
Python
FastAPI
Pydantic
PyJWT
SERVICE
Planner Agent
Central orchestrator using LangGraph to manage the stateful
Python
LangGraph
LangChain
Google Gemini
Specialized Agent Swarm
SERVICE
Vision Agent
Specialized agent for analyzing images and videos using Gemini
Python
Google Gemini Vision
OpenCV
SERVICE
Knowledge Agent
RAG-enabled agent for retrieving municipal policies and government
Python
ChromaDB
Sentence-Transformers
LangChain
SERVICE
Location Agent
Agent for reverse geocoding and ward identification using
Python
GeoPy
Model Context Protocol (MCP)
SERVICE
Routing Agent
Logic agent to determine department routing and SLA priority.
Python
Google Gemini
LangChain
SERVICE
Complaint Agent
Agent responsible for CRUD operations on complaints and
Python
SQLAlchemy
Model Context Protocol (MCP)
SERVICE
Notification Agent
Agent for sending multi-channel notifications (Email, SMS, Push).
Python
SendGrid API
Firebase Cloud Messaging
MCP
SERVICE
Analytics Agent
Agent for generating real-time heatmaps and performance metrics.
Python
Pandas
Matplotlib
SQLite
Vector Search
Integration & LLM Services
SQL Query
SERVICE
MCP Server
Standardized interface for agents to access external tools and
Python
MCP SDK
SQLite
GSM API
SERVICE
Gemini API
Analytics Read
der for reasoning, vision, and text generation.
Google Gemini 1.5 Pro
Google AI Studio
Persistence Layer
DATABASE
CivicFlow Main DB
Stores core data
CivicFlow Main DB
DATABASE
Policy Vector DB
Stores core data
Policy Vector DB